

# Causal Effect of Resource Sustainability Act on Targeted Waste Categories in Singapore

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## Abstract

This paper evaluates the causal effect of Singapore’s Resource Sustainability Act (RSA) on recycling rates across six targeted waste categories using a staggered Difference-in-Differences framework applied to a balanced panel of eight waste streams from 2000 to 2024. The RSA was introduced in three waves: packaging materials (Paper, Plastic, Glass) in 2020; e-waste streams (Ferrous and Non-Ferrous Metals) in 2021; and Food waste in 2024. I estimated two-way fixed effects (category and year wise) staggered difference-in-difference model, clustering standard errors at the category level, and validated the parallel trends assumption through in-time placebo tests and pre-trend diagnostics. Wave 1 packaging categories experienced a decline of 0.47 log points relative to never-treated controls following the 2020 RSA rollout. Wave 2 metal categories show no measurable change. Food waste is analysed via synthetic control due to a pre-trend violation; the post-2024 gap suggests a positive effect, pending further data. These findings indicate that mandatory reporting initially reduced measured recycling performance for packaging categories, reflecting short-term adjustment costs as firms restructured collection and reporting infrastructure under the new regulatory framework.

**Keywords:** Waste Management, Resource Sustainability Act, Recycling Rates, Staggered Difference-in-Differences, Synthetic Control

## 1 Introduction

Can mandatory producer reporting requirements reduce packaging waste, or does regulation merely shift compliance reporting without changing behavior? This question sits at the intersection of environmental regulation and firm behaviour, and it is the central empirical question this paper addresses using Singapore’s Resource Sustainability Act (RSA) as a natural experiment. The RSA, enacted in 2020, introduced extended producer responsibility (EPR) schemes with mandatory packaging reporting, e-waste management, and food waste segregation requirements (Ministry of Sustainability and the Environment, 2020). Its staggered rollout across three waves, packaging in

2020, e-waste in 2021, food waste in 2024, provides the identifying variation I exploit for a causal difference-in-difference design.

Singapore’s environmental governance has evolved from a primary focus on basic sanitation and public health in the 1970s to strong emphasis on circular economy initiatives and resource efficiency (Ministry of Environment and Water Resources, 2019a). This shift is driven by land scarcity: the Semakau Landfill, the nation’s sole disposal site, is projected to reach capacity by 2035. To address this, Singapore has historically relied on waste-to-energy incineration, with facilities processing approximately 7,600 tonnes of waste daily (Ministry of Environment and Water Resources, 2019a). New incineration plants are required every 7-10 years to keep pace with waste volumes, making this reliance both financially and environmentally unsustainable. In response, the government introduced the Zero Waste Masterplan in 2019, setting targets to reduce daily waste sent to Semakau by 30% by 2030 and achieve a 70% overall recycling rate, with sub-targets of 81% for non-domestic and 30% for domestic waste (Ministry of Environment and Water Resources, 2019b). As exporting waste is not a viable long-term solution, Singapore is augmenting its infrastructure with the Tuas Nexus incineration plant, scheduled for completion around 2027 (National Environment Agency, 2025), to ensure long-term waste management resilience.

Recycling performance varies dramatically across categories: construction and demolition materials, and ferrous/non-ferrous metal streams achieve near-100% recycling rates, while food waste, e-waste, and packaging waste remain persistently low (Ministry of Sustainability and the Environment, 2025a). Reducing food waste at the household level is constantly addressed through public campaigns and reminders, but progress remains limited (Ministry of Sustainability and the Environment, 2025b).

Figure 1 illustrates these divergent trajectories and marks the three RSA implementation waves. The RSA represents the most significant upstream regulatory intervention in Singapore’s waste history, targeting waste reduction and resource recovery before waste enters the disposal system.

The RSA introduces three key regulatory mechanisms. First, it establishes an EPR framework for e-waste, under which NEA appointed ALBA E-Waste Smart Recycling Pte Ltd to exclusively handle regulated consumer e-waste from July 2021 to June 2026, executing collection, treatment, and recycling obligations for all producers (Allen & Gledhill, 2021; National Environment Agency, 2026). E-waste streams contain ferrous metals, non-ferrous metals, glass, and plastics (Logakanthi, 2017), making the EPR framework directly relevant to Wave 2 categories in this study. Second, it mandates packaging reporting: producers with annual turnover exceeding \$10 million must register with NEA and submit 3R plans detailing packaging use and reduction targets, with the first submission due March 31, 2022 (Begum & Lun, 2025). Third, for the owners and operators of commercial and industrial premises i.e. large hotels, malls, large industrial developments housing food manufacturers, and food caterers, segregation of food waste for treatment from March 2024 is mandatory (National Environment Agency, 2020b). In addition, new buildings constructed after 2021 must allocate space for on-site food waste treatment in their design. Following completion

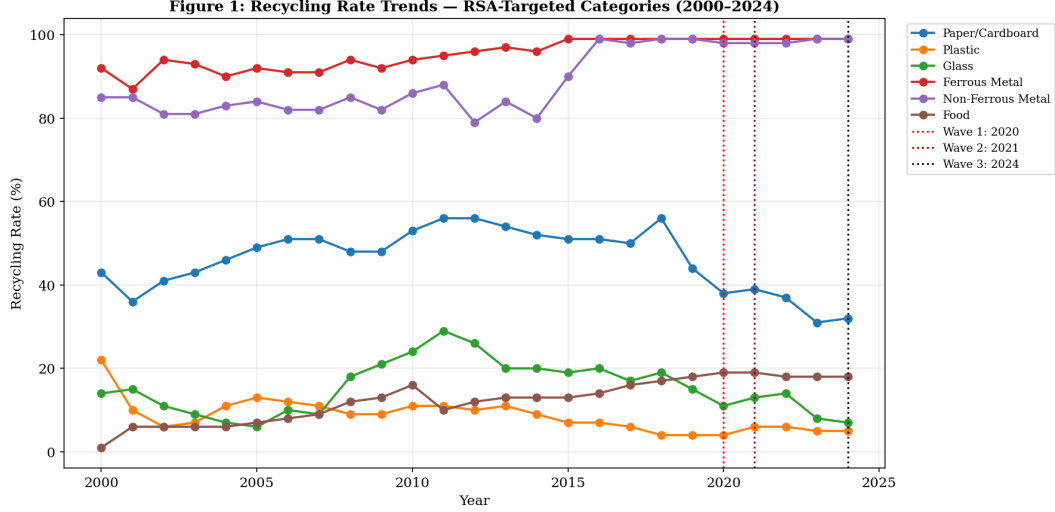


Figure 1: Annual Recycling Rates and RSA Implementation Waves  
Notes: Annual recycling rates (%) for eight waste categories (2000–2024). Dotted lines mark RSA Wave 1 (2020), Wave 2 (2021), and Wave 3 (2024). The performance gap between metals and other categories motivates heterogeneous analysis. Data: MSE & NEA (2026).

of the Tuas Nexus facility, compliance will extend to newly established food industries. These mechanisms operate upstream of the waste stream, targeting firms, which is the scope of my study.

Understanding whether such regulatory frameworks actually change recycling outcomes, rather than just changing what firms report, is an important empirical question. A growing body of literature evaluates environmental policies using quasi-experimental methods, particularly difference-in-differences approaches that exploit staggered implementation. At the household level, Alacevich et al. (2021) demonstrated that introducing organic waste sorting in a Swedish municipality (Partille) significantly reduced household waste generation, with spillover effects that encouraged short-term reductions in overall waste, though these effects diminished over time. They used staggered DiD on a panel of 4,300 households spanning 2012–2017, a method I utilize in my study; their cost-benefit analysis found the policy took 5.5 years to pay off. Building on such localised evidence on waste sorting, many countries have adopted EPR frameworks at a systemic level. Within Europe, the German Packaging Act (*Verpackungsgesetz*), implemented in 2019, targeted similar categories (glass, plastic, paper, ferrous metals, aluminium) by introducing mandatory registration, stricter reporting, and higher recycling targets across packaging materials. Pruess and Garrett (2025) evaluate EPR effectiveness for plastic packaging across Belgium, France, and Germany, finding that full financial coverage is key for long-run compliance, while shared challenges including enforcement gaps and free-riding from unclear producer definitions persist across systems. Lin and Rijal (2024) examine how regulatory design varies in its effect on waste sorting across countries by evaluating the eco-efficiency of decision-making units (DMUs). My paper contributes to the ex-post causal evaluation of a national policy RSA in Singapore.

The paper proceeds as follows. Section 2 describes the data and sample selection. Section 3 presents

the empirical strategy, including the staggered DiD specification, identification assumptions, and robustness approach. Section 4 reports results: descriptive statistics, parallel trends validation, main DiD estimates, COVID-19 robustness, event study dynamics, Callaway–Sant’Anna group-time estimates, and the synthetic control for Food. Section 5 concludes.

## 2 Data and Sample Selection

### 2.1 Data Source

I use a balanced panel of annual waste management data published by Singapore’s Ministry of Sustainability and the Environment and the National Environment Agency (Ministry of Sustainability and the Environment & National Environment Agency, 2026). The dataset covers 14 waste categories from 2000 to 2024 (25 years), recording annual tonnes generated, disposed, and recycled for each category. I construct the recycling rate as recycled tonnes divided by generated tonnes, expressed as a percentage. After conducting pre-diagnostic tests, certain variables are dropped due to their volatility. The dataset used for analysis contains 200 observations across 8 categories, after filtering out missing values.

### 2.2 Sample Selection and Variable Dropping Justification

The original dataset contains 14 waste categories. I dropped six categories for the following reasons:

*Ash and Sludge* is excluded because it is an incineration by-product. Its volume is determined by the total quantity of waste incinerated across all categories. Including it would introduce a structural endogeneity problem, as its “recycling rate” responds to inputs from other categories rather than to specific category that RSA mandates on. *Others* is also dropped because it is a residual category containing an undefined and time-varying mix of waste types that I cannot consistently track or map to specific RSA targeted or untargeted variables.

*Wood*, *Textile/Leather*, *Horticultural waste*, and *Scrap Tyres* are not considered for the analysis after conducting pre-trend diagnostic testing. These categories exhibit high within-category volatility in pre-treatment recycling rates (coefficients of variation exceeding 0.40) and fail formal pre-trend slope comparisons relative to the control group. In other words, excluding these categories ensures a cleaner control group by removing unrelated dynamics that could bias the results. Sample selection follows standard practice by including only units with stable pre-treatment trends, ensuring the donor pool provides a valid counterfactual (Abadie et al., 2010).

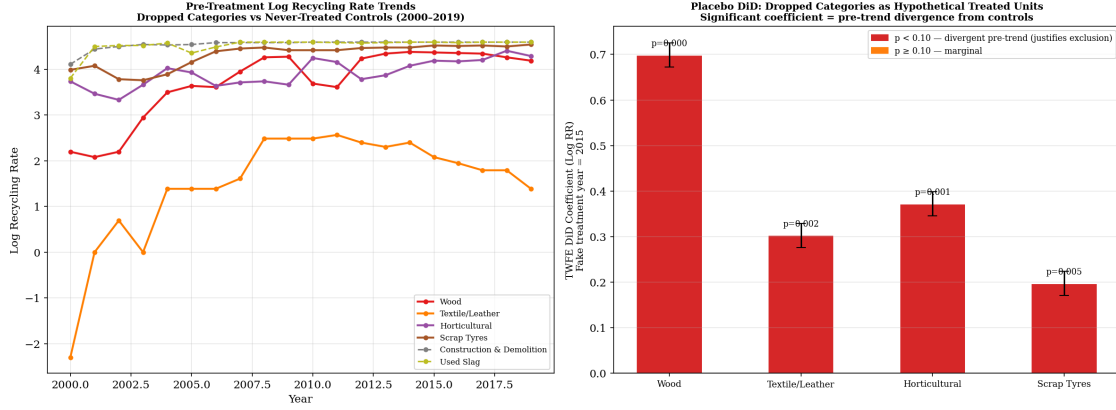


Figure 2: Pre-treatment Trends and Placebo DiD for Excluded Categories

*Notes:* Left panel: log recycling rates for categories excluded due to divergent trends. Right panel: Placebo DiD coefficients using these categories as hypothetical treated units. Significant coefficients ( $p < 0.10$ ) confirm that these categories violate the parallel trends assumption relative to the control group.

The retained sample consists of three RSA-targeted packaging categories (Paper/Cardboard, Plastic, Glass), two RSA-targeted metal categories (Ferrous Metal, Non-Ferrous Metal), one food waste category (Food), and two never-treated control categories (Construction & Demolition, Used Slag). The RSA-targeted categories entered treatment in a staggered fashion: Packaging in Wave 1 (2020), E-waste in Wave 2 (2021), and Food waste in Wave 3 (2024).

### 2.3 Outcome Variable and Transformation

The primary outcome is  $\log(\text{recycling rate})$ , where I replace zero values with 0.1 before taking logs. The log transformation is needed on three grounds. First, recycling rate levels differ dramatically across categories; and log-transforming compresses this scale difference without distorting within-category dynamics, as shown in Figure 3. It displays pre-treatment recycling rate trends for treated and control categories separately by wave, in both levels and log scale. In levels (left column), the structural difference in scale between packaging categories ( $\sim 10\text{-}50\%$ ) and controls ( $\sim 95\text{-}99\%$ ) is apparent. In log scale (right column), the pre-treatment trajectories align substantially, with log-transformed slopes clustered between  $-0.043$  and  $+0.042$  across all categories. The vertical dashed line at 2015 marks the fake treatment date; no discontinuity is visible at this point, consistent with the statistically insignificant placebo coefficients reported in Table 1. The correlation matrix in Figure 7 further illustrates this: Plastic shows a strong negative correlation with both control categories ( $r = -0.71, -0.78$ ), reflecting its diverging level trajectory, while Wave 2 ferrous and non-ferrous metals show moderate positive correlations consistent with their ceiling-level dynamics. Third, placebo tests using a fake 2015 treatment date pass for all retained waste categories, tested under log transformation.

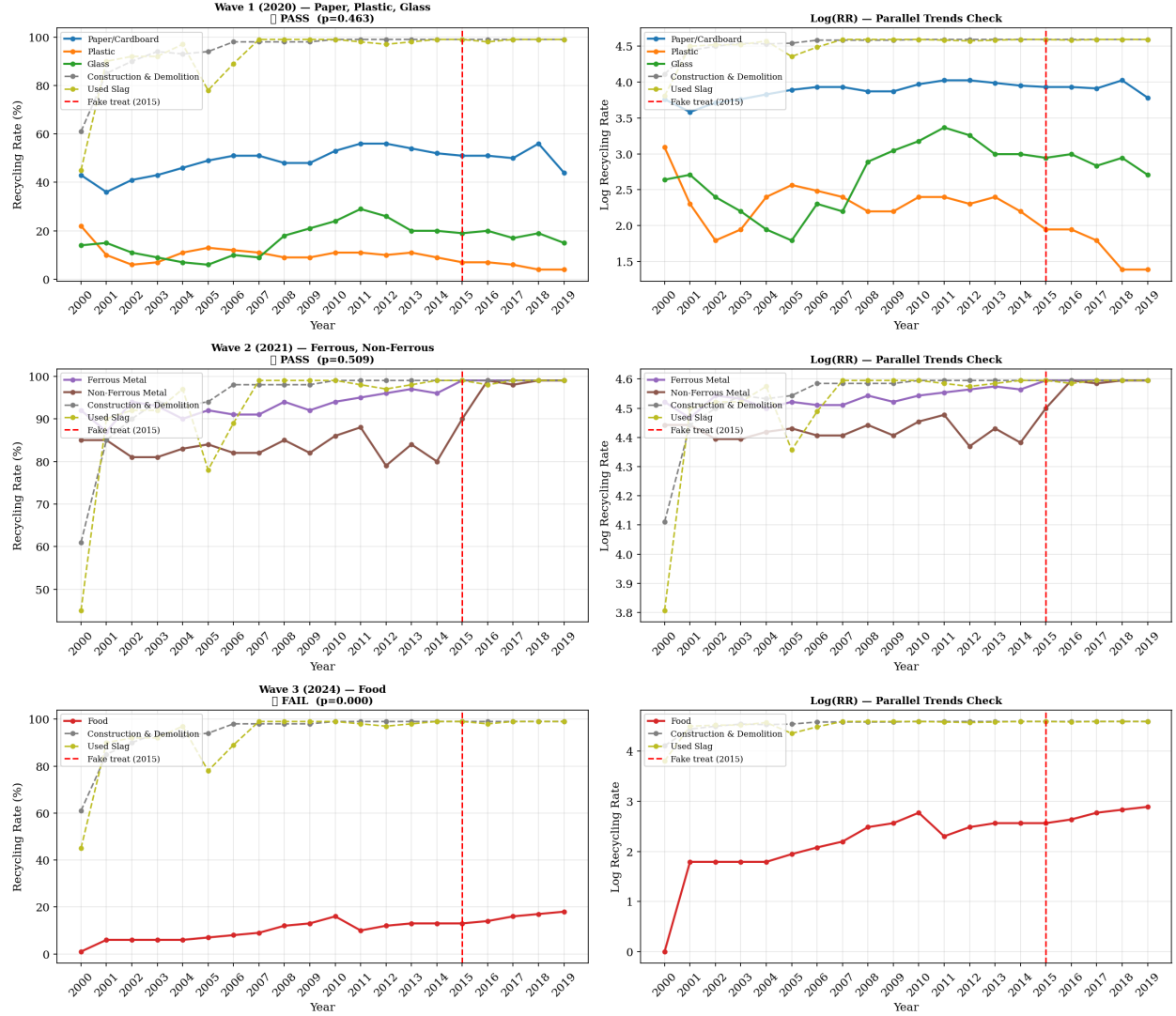


Figure 3: Parallel Trends Validation: Recycling Rates by Treatment Wave

*Notes:* Comparison of recycling rates (left) and log recycling rates (right) for treatment cohorts against the control pool (Construction & Demolition and Used Slag). Vertical dashed lines mark the placebo treatment year (2015). A 'PASS' indicates no statistically significant difference in pre-treatment trends, justifying the use of the control pool for these cohorts; 'FAIL' indicates divergent trends necessitating alternative methods i.e., Synthetic Control.

### 3 Empirical Strategy

#### 3.1 Staggered DiD Specification

I estimate a two-way fixed effects staggered DiD model. The baseline specification is:

$$\log(Y_{it}) = \sum_{w \in \{1,2\}} \beta_w \cdot D_{iw} \cdot \mathbf{1}[t \geq g_w] + \alpha_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where  $Y_{it}$  is the recycling rate of waste category  $i$  in year  $t$ ;  $D_{iw}$  is an indicator equal to one if category  $i$  belongs to treatment wave  $w$ ;  $g_w$  is the first treated year for wave  $w$  (2020 for Wave 1, 2021 for Wave 2);  $\alpha_i$  are waste-category fixed effects that absorb all time-invariant differences across categories;  $\lambda_t$  are year fixed effects that absorb common macroeconomic shocks and time trends; and  $\varepsilon_{it}$  is an idiosyncratic error term. I cluster standard errors at the waste-category level to account for serial correlation within categories over time.

### 3.2 Identification Assumptions

The causal interpretation of  $\hat{\beta}_w$  rests on the conditional parallel trends assumption. Absent the RSA, treated and control categories would have followed parallel trends in recycling rates (logarithm form), after controlling for category and year fixed effects. I evaluate this assumption formally through event study as shown in Figure 4. I also conduct placebo test and 500 random permutation test.

I restrict the sample to pre-2020 data and re-estimate Equation 1. Under the null of parallel trends, the placebo DiD coefficient should be statistically indistinguishable from zero. The Wave 1 placebo coefficient is  $-0.204$  ( $p = 0.449$ ) and Wave 2 placebo coefficient is  $0.0306$  ( $p = 0.492$ ) far from conventional significance thresholds. This strongly supports the parallel trends assumption for both treated cohorts. These results are shown in Table 1. I exclude Food waste (Wave 3) from the DiD framework because it violates the parallel trends assumption. I address it via synthetic control in Section 4.6. The Fisher permutation test is underpowered in this setting: five treatment units generate too few distinct permutations to construct a reliable null distribution, and the resulting  $p$ -value should be interpreted with caution.

Table 1: Placebo Tests: Parallel Trends Validation.

| Test                                                  | Coefficient | $p$ -value | Result                    |
|-------------------------------------------------------|-------------|------------|---------------------------|
| P1: Fake year (2015) — Wave 1 (Paper, Plastic, Glass) | $-0.2044$   | 0.449      | Pass                      |
| P1: Fake year (2015) — Wave 2 (Ferrous, Non-Ferrous)  | $0.0306$    | 0.492      | Pass                      |
| P2: Permutation / Fisher ( $n = 500$ )                | $-0.2723$   | 0.400      | Underpowered <sup>†</sup> |

*Notes:* Outcome:  $\log(\text{recycling rate})$ . Two-way FE. Cluster-robust SE by waste type. <sup>†</sup>Low power with five units. Placebo 1 assigns a fictitious treatment year (2015) to each wave and tests for a spurious DiD coefficient in pre-2020 data; a  $p$ -value above 0.10 constitutes a pass. Permutation test result is underpowered given five treatment units and should be interpreted cautiously.

### 3.3 Justification for use of Standard TWFE Estimator

Under staggered adoption, standard TWFE implicitly uses already-treated units as controls for later-treated cohorts, assigning negative weights for  $2 \times 2$  DiD comparisons (Goodman-Bacon, 2021;

Sun & Abraham, 2021; Callaway & Sant’Anna, 2021). This matters because negative weights can reverse the sign of the estimated effect entirely, a policy that genuinely reduced recycling rates could appear to have raised them. The bias is worst when treated cohorts experience heterogeneous effects and when early-treated units partially recover relative to later-treated ones.

This study faces limited exposure to that problem for two reasons. First, the comparison group consists exclusively of never-treated categories (Construction & Demolition and Used Slag). Negative weighting only arises when already-treated units are used as controls; by restricting comparisons to never-treated units throughout, this source of bias is eliminated by design. Second, there are only two treated cohorts separated by one year, which limits the number of periods in which Wave 1 could contaminate Wave 2 estimates. As a direct check, I compute Callaway–Sant’Anna  $ATT(g, t)$  estimates using the never-treated comparator (Callaway & Sant’Anna, 2021). These estimates, reported in Section 4.5 along with the Figure 5, align closely with the TWFE results, confirming the main findings are not an artefact of negative weighting.

### 3.4 COVID-19 and RSA Confounding

A central identification threat is that the RSA Wave 1 rollout in 2020 coincides precisely with the COVID-19 pandemic. Lockdowns reduced commercial activity, altered packaging consumption patterns, and disrupted waste collection operations. If pandemic-related shocks differentially affected packaging categories relative to construction and used slag, the estimated DiD coefficient could reflect transient COVID disruptions rather than the impact of the RSA.

Because 2020 and 2021 represent the primary implementation phase for the RSA for packaging and e-waste, these years are essential to the identification of the policy effect. I cannot precisely measure the impact of lockdown-specific shifts due to the lack of monthly data. Stable pre-trends indicate that staggered DiD approach can be considered but the estimates should be interpreted with caution.

## 4 Results

### 4.1 Descriptive Statistics

Table 2 presents descriptive statistics. Wave 1 packaging categories (Paper, Plastic, Glass) have a mean pre-treatment recycling rate of 23.39%, far below the control group average of 95.04%. Wave 2 metal categories average 92.16%, close to the control mean, reflecting that these streams were already highly recycled before the RSA, which is confirmed through Figure 1. The near-equality of pre-RSA and post-RSA mean recycling rates for all treated categories (51.19% vs. 49.72%) masks substantial heterogeneity across waves, which the DiD estimates decompose. Figure 8 confirms that the aggregate recycled share remained broadly stable over 2000-2024, reinforcing that any



category-level changes identified in the DiD are not driven by aggregate waste volume shifts.

Table 2: Descriptive Statistics: Recycling Rates by Category Group (2000-2024).

|                                          | Obs | Mean  | Std. Dev. | Min  | Median | Max  |
|------------------------------------------|-----|-------|-----------|------|--------|------|
| <i>RSA-Targeted Categories (DiD)</i>     |     |       |           |      |        |      |
| Wave 1 (2020) — Paper, Plastic, Glass    | 75  | 23.39 | 17.55     | 4.0  | 15.0   | 56.0 |
| Wave 2 (2021) — Ferrous, Non-Ferrous     | 50  | 92.16 | 6.78      | 79.0 | 94.0   | 99.0 |
| All Treated (Waves 1+2)                  | 125 | 50.90 | 36.69     | 4.0  | 48.0   | 99.0 |
| <i>Synthetic Control</i>                 |     |       |           |      |        |      |
| Wave 3 (2024) — Food                     | 25  | 12.32 | 5.10      | 1.0  | 13.0   | 19.0 |
| <i>Never-Treated Control</i>             |     |       |           |      |        |      |
| Construction & Demolition, Used Slag     | 50  | 95.04 | 9.77      | 45.0 | 99.0   | 99.0 |
| <i>Pre/Post Comparison — DiD Treated</i> |     |       |           |      |        |      |
| Pre-RSA (2000-2019)                      | 100 | 51.19 | 35.42     | 4.0  | 50.5   | 99.0 |
| Post-RSA (2020-2024)                     | 25  | 49.72 | 42.17     | 4.0  | 37.0   | 99.0 |
| Full Sample                              | 200 | 57.11 | 38.77     | 1.0  | 56.0   | 99.0 |

*Notes:* Balanced panel. 8 categories  $\times$  25 years = 200 observations. Recycling rates in percent. Recycling rate is annual recycled tonnes divided by generated tonnes, expressed as a percentage. The sample covers eight waste categories over 25 years (200 observations). Pre-RSA refers to years 2000-2019; Post-RSA to 2020-2024. I exclude Food from the DiD treated group and analyse it separately via synthetic control.

## 4.2 Main DiD Results

Table 3 presents the main DiD estimates. Column (1) pools all treated categories (Waves 1 and 2) against the never-treated controls. The coefficient of  $-0.2723$  ( $p = 0.074$ ) implies that treated categories experienced approximately a 24% relative decline in recycling rates post-RSA compared to controls. This pooled estimate masks substantial heterogeneity across waves.

Column (2) restricts the sample to Wave 1 packaging categories. The estimated effect is  $-0.475$  ( $p = 0.001$ ), implying that Paper, Plastic, and Glass recycling rates fell by approximately 38% relative to the never-treated categories following the 2020 RSA implementation. Such a decline suggests that firms and collection operators had to restructure waste segregation, mandatory reporting, and downstream processing arrangements simultaneously in the initial phase of RSA implementation.

Column (3) shows that Wave 2 metal categories (Ferrous and Non-Ferrous) exhibit no statistically significant RSA effect ( $\hat{\beta} = 0.0312$ ,  $p = 0.424$ ). This null result is informative rather than a failure of identification: both categories had recycling rates at or above 99% before 2021. A category already recycling 99% of its waste stream has no room to improve under any regulatory intervention, through practices like producer registration and reporting obligations. The RSA changed administrative compliance while keeping the trend of ferrous and non-ferrous metals almost the same.

Column (4) provides a levels robustness check using the raw recycling rate as the outcome. The point estimate of  $-6.17$  percentage points is negative and directionally consistent with the log specification, but is not significant ( $p = 0.200$ ). The loss of precision in levels confirms that the log specification is an appropriate choice: the levels estimator is sensitive to the large scale gap between packaging categories (9-56%) and metal categories (79-99%), compressing what is a real within-category effect into a noisy cross-category comparison.

Table 3: Main DiD Estimates: Effect of RSA on Recycling Rate.

|                 | (1)         | (2)             | (3)      | (4)         |
|-----------------|-------------|-----------------|----------|-------------|
|                 | All Treated | Wave 1          | Wave 2   | All Treated |
|                 | Log(RR)     | Log(RR)         | Log(RR)  | Levels (pp) |
| DiD coefficient | $-0.2723^*$ | $-0.4749^{***}$ | 0.0312   | $-6.1700$   |
|                 | (0.1258)    | (0.0590)        | (0.0338) | (4.2825)    |
| $p$ -value      | 0.074       | 0.001           | 0.424    | 0.200       |
| Observations    | 175         | 125             | 100      | 175         |
| Clusters        | 7           | 5               | 4        | 7           |
| Waste-type FE   | Yes         | Yes             | Yes      | Yes         |
| Year FE         | Yes         | Yes             | Yes      | Yes         |

Notes:  $***p < 0.01$ ,  $**p < 0.05$ ,  $*p < 0.10$ . Cluster-robust SE in parentheses. Outcome:  $\log(\text{recycling rate})$ . Each column reports the DiD coefficient from a two-way fixed effects regression with waste-type and year fixed effects. I cluster standard errors by waste type (in parentheses). Column (1) pools all DiD-treated categories (Waves 1 and 2); Columns (2) and (3) restrict to individual waves; Column (4) uses raw recycling rate (percentage points) as a levels robustness check. The treatment variable is  $\text{Post}_t \times \text{Treated}_i$ , where  $\text{Post}_t$  is defined relative to each wave's implementation year. Never-treated baseline: Construction & Demolition and Used Slag.

### 4.3 COVID-19 Robustness

Table 4 presents three robustness specifications designed to isolate the RSA effect from COVID-19 confounding. When I exclude 2020-2021 from the sample entirely (Specification R2), the coefficient rises to  $-0.2965$  ( $p = 0.068$ ), stronger than the baseline of  $-0.2723$ . The fact that removing the pandemic years 2020 and 2021 strengthens rather than weakens the estimate is the key identifying evidence: if the 2020 and 2021 recycling decline reflected COVID disruption rather than RSA implementation, removing those years would pull the coefficient towards zero. Instead, the structural decline in packaging recycling rates persists ruling out pandemic confounding. These results should nonetheless be interpreted with caution due to the lack of monthly data.

Excluding only 2020 (Specification R3) produces  $-0.2664$  ( $p = 0.070$ ), consistent with the baseline. This is consistent with firms requiring time to adjust in the initial year of RSA implementation (2020). Across all three specifications, the sign, magnitude, and significance of the RSA effect are

stable.

Table 4: COVID-19 Robustness Checks: DiD Estimates of RSA Effect on Log Recycling Rate.

| Spec. | Sample                          | Coefficient | Std. Error | $p$ -value |
|-------|---------------------------------|-------------|------------|------------|
| R1    | Baseline: full sample 2000-2024 | $-0.2723^*$ | (0.1258)   | 0.074      |
| R2    | Exclude 2020-2021 entirely      | $-0.2965^*$ | (0.1334)   | 0.068      |
| R3    | Exclude 2020 only               | $-0.2664^*$ | (0.1210)   | 0.070      |

Notes:  $***p < 0.01$ ,  $**p < 0.05$ ,  $*p < 0.10$ . Two-way FE. Cluster-robust SE. Outcome:  $\log(\text{recycling rate})$ . I re-estimate the main DiD specification under three different treatments of the COVID-19 period (2020-2021). Specifications R1-R3 use all treated categories (Waves 1 and 2). Two-way Fixed Effects throughout; I cluster standard error by waste type.

#### 4.4 Dynamic DiD: Event Study

Figure 4 presents event study estimates for Waves 1 and 2, plotting the DiD coefficient at each year relative to treatment. The baseline period is  $t = -1$ . Under parallel trends, pre-treatment coefficients should cluster around zero. For Wave 1, pre-treatment coefficients at  $t - 2$  and  $t - 3$  are positive and statistically insignificant at the 5% level. The marginally significant values at  $t - 4$  and  $t - 5$  reflect the pre-existing structural decline in Paper/Cardboard recycling rates from 2015 onward visible in Figure 1 driven by factors other than RSA-specific anticipatory behaviour. This pre-trend begins five years before RSA enactment and precedes all legislative procedures.

The two pieces of evidence i.e., in-time placebo test and COVID robustness checks confirm it does not invalidate the identification strategy. The in-time placebo test assigns a fictitious 2015 treatment date and finds no spurious effect (Table 1,  $p = 0.449$ ), confirming that the pre-2020 trend does not mimic a treatment effect. Second, the COVID robustness checks show the post-treatment effect strengthens when 2020-2021 are excluded entirely (R2:  $-0.2965$ ) and remains stable when only 2020 is excluded (R3:  $-0.2664$ ). The smooth transition through  $t = 0$  indicates that firms responded to the RSA only upon implementation, satisfying the no-anticipation assumption required for causal identification. Post-treatment, the coefficient turns negative from  $t = 0$  and grows in magnitude through  $t + 3$  and  $t + 4$  ( $-0.2468$  and  $-0.2807$  respectively). Firms took approximately three to four years to restructure their packaging waste infrastructure under the new reporting regime.

For Wave 2, event study coefficients are near-zero throughout the entire pre- and post-treatment window, confirming the null result from the static specification. The flat trajectory indicates that EPR obligations for metal categories formalised existing recycling practice without altering behaviour.

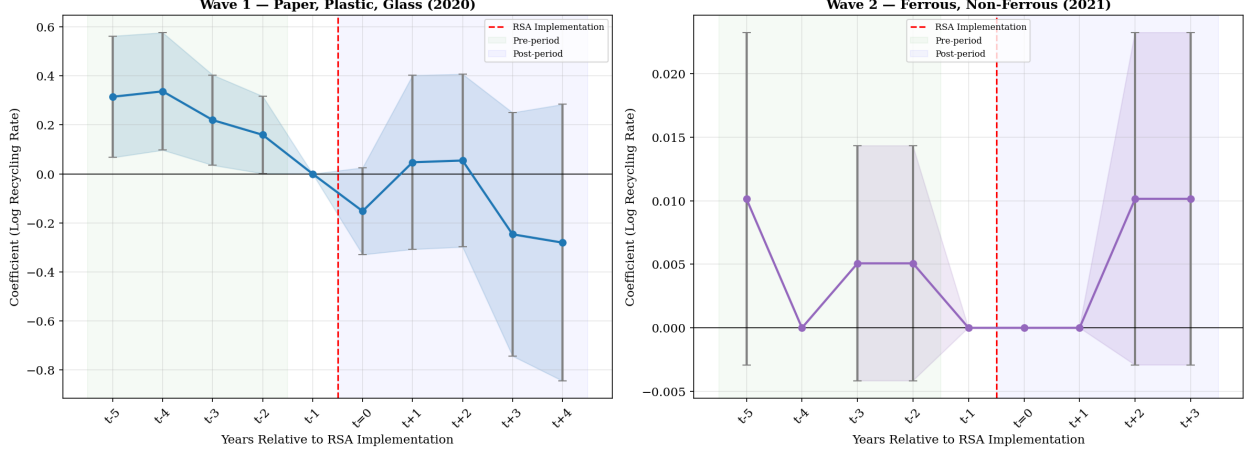


Figure 4: Event Study: Dynamic RSA Effects by Treatment Wave.

*Notes:* Y-axis: two-way fixed effects DiD coefficient in log recycling rate units. X-axis: years relative to RSA implementation ( $t = -1$  is the omitted baseline, normalised to zero). Shaded bands: 95% confidence intervals. The vertical red dashed line marks  $t = 0$ , the RSA implementation year. Points to the *left* of the dashed line are pre-treatment estimates clustering near zero confirms treated and control categories did not diverge before the RSA. Points to the *right* show how far packaging and e-waste recycling fell relative to the counterfactual in each post-treatment year. *Left panel:* Wave 1 packaging categories (Paper, Plastic, Glass) vs. never-treated controls, 2015–2024. *Right panel:* Wave 2 metal categories (Ferrous, Non-Ferrous) vs. never-treated controls, 2016–2024. The flat right panel confirms the ceiling-effect null result for metals.

#### 4.5 Callaway-Sant’Anna Group-Time ATT Estimates

To address the negative weighting concern, I compute Callaway-Sant’Anna  $ATT(g, t)$  estimates using the never-treated comparator (Callaway & Sant’Anna, 2021). I construct these as clean  $2 \times 2$  DiD estimates for each cohort  $g$  and post-treatment period  $t$ , relative to the  $t = -1$  baseline, which eliminates contamination from already-treated units. This is shown in Table 5.

The cohort-2020 estimates provide period-by-period confirmation of the static result: the effect is  $-0.1523$  during the event ( $t = 0$ ), turns slightly positive at  $t + 1$  ( $+0.0472$ ) and  $t + 2$  ( $+0.0544$ ) as firms begin to adapt to the new reporting regime, then deepens sharply to  $-0.2468$  and  $-0.2807$  at  $t + 3$  and  $t + 4$ . The mean ATT for the 2020 cohort across all post-treatment periods is  $-0.1156$  log points equivalent to an 11% relative reduction in recycling rates ( $e^{-0.1156} - 1 \approx -0.109$ ). Three features of this trajectory are economically meaningful. The initial impact at  $t = 0$  is negative but moderate ( $-0.1523$ ), reflecting immediate disruption to existing collection and reporting arrangements. The partial recovery at  $t + 1$  and  $t + 2$  indicates that firms made genuine initial compliance investments, they began adapting their infrastructure in response to the mandatory reporting requirements. The subsequent deepening to  $-0.2468$  and  $-0.2807$  at  $t + 3$  and  $t + 4$  reveals that the structural demands of the full mandatory packaging reporting and segregation system proved more complex and costly than firms initially anticipated, causing recycling volumes to fall further as the compliance burden accumulated. For the 2021 cohort, all ATT estimates are at or near zero (mean:  $+0.0051$ ), confirming the ceiling-effect finding: metal categories at 99% recycling rates have

no behavioural margin to adjust regardless of regulatory pressure. Figure 5 plots the  $ATT(g,t)$  trajectory for both cohorts; the full set of period-by-period estimates is reported in Table 5. The consistency between TWFE and C&S estimates confirms that negative weighting does not distort the main conclusions.

Standard errors and  $p$ -values in Table 5 are not reported, as inference-valid bootstrap standard errors require implementation via the DiD R package; point estimates are reported for qualitative comparison with TWFE results only.

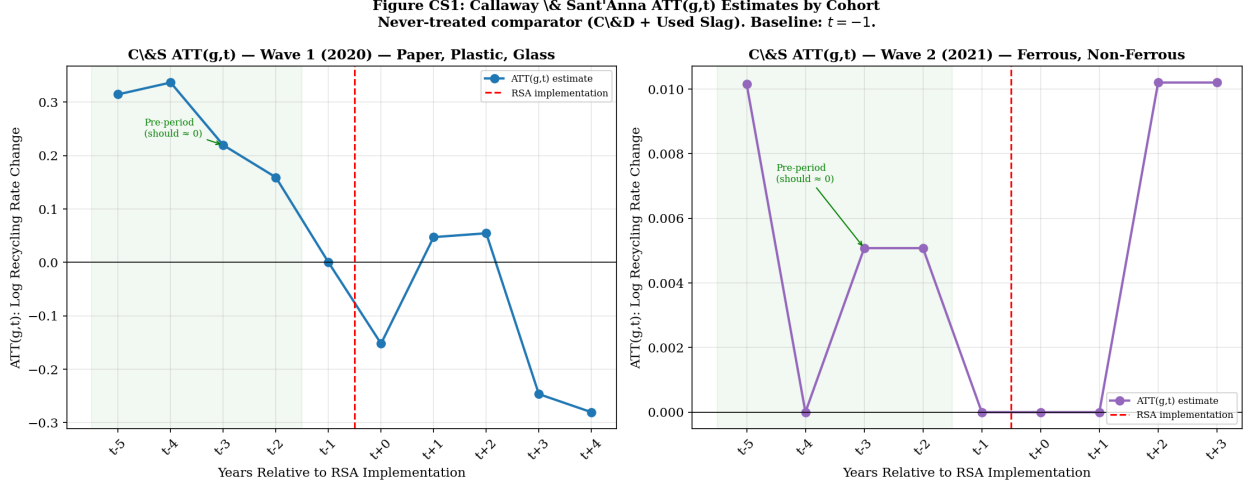


Figure 5: Callaway-Sant'Anna  $ATT(g,t)$  Estimates by Cohort.

*Notes:* Y-axis:  $ATT(g,t)$  in log recycling rate units. X-axis: years relative to RSA implementation for each cohort ( $t = -1$  is the omitted baseline). Unit of observation: waste category-year. Each point is a clean  $2 \times 2$  DiD estimate comparing cohort  $g$  to never-treated controls (Construction & Demolition and Used Slag) in period  $t$ , with no contamination from already-treated units. Points to the *left* of the dashed red line are pre-treatment estimates; their proximity to zero confirms parallel trends hold cohort-by-cohort, independently of the TWFE model. *Left panel:* Cohort 2020 (Wave 1: Paper, Plastic, Glass). *Right panel:* Cohort 2021 (Wave 2: Ferrous, Non-Ferrous). The U-shaped trajectory in the left panel is the same pattern visible in the event study and is confirmed here to be free from negative weighting bias.

Table 5: Callaway-Sant’Anna  $ATT(g, t)$  Estimates by Cohort and Period.

| Cohort $g$            | Period $t$ | Rel. Time | $ATT(g, t)$ | Wave   |
|-----------------------|------------|-----------|-------------|--------|
| 2020                  | 2020       | $t + 0$   | -0.1523     | Wave 1 |
| 2020                  | 2021       | $t + 1$   | +0.0472     | Wave 1 |
| 2020                  | 2022       | $t + 2$   | +0.0544     | Wave 1 |
| 2020                  | 2023       | $t + 3$   | -0.2468     | Wave 1 |
| 2020                  | 2024       | $t + 4$   | -0.2807     | Wave 1 |
| 2021                  | 2021       | $t + 0$   | 0.0000      | Wave 2 |
| 2021                  | 2022       | $t + 1$   | 0.0000      | Wave 2 |
| 2021                  | 2023       | $t + 2$   | +0.0102     | Wave 2 |
| 2021                  | 2024       | $t + 3$   | +0.0102     | Wave 2 |
| Mean ATT: Cohort 2020 |            |           | -0.1156     |        |
| Mean ATT: Cohort 2021 |            |           | +0.0051     |        |

Notes: Outcome:  $\log(\text{recycling rate})$ . Never-treated categories for comparison: Construction & Demolition and Used Slag. All estimates are in log-point units.  $ATT(g, t)$  is the Average Treatment Effect on the Treated for cohort  $g$  at time  $t$ , computed as a clean  $2 \times 2$  DiD relative to never-treated controls and the baseline is  $t = -1$ . Relative time equals  $t - g$ .

#### 4.6 Synthetic Control: Food Waste (Wave 3)

I did not analyse Food waste (Wave 3, treated in 2024) within the DiD framework because it violates the parallel trends assumption. Food’s pre-treatment log recycling rate increases at a slope of +0.088 log points per year, compared to +0.011 and +0.016 for Construction & Demolition and Used Slag respectively. This structural divergence reflects independent secular trends driven by pre-existing NEA food waste reduction programmes. The divergence generates a statistically significant placebo coefficient for every fake treatment year tested (2005-2018), confirming that DiD identification is invalid for this category.

Instead, I apply the synthetic control method (Abadie et al., 2010, 2015) and construct a weighted combination of donor categories that best replicates Food’s pre-treatment log recycling trajectory. I select the optimal donor weights by minimising pre-treatment mean squared prediction error (MSPE). The solution places 85.6% weight on Glass and 14.4% on Plastic, producing a pre-treatment RMSPE of 0.714 log points. The pre-treatment RMSPE reflects the inherent difficulty of matching Food’s distinctive secular upward trajectory using any weighted combination of the available donor categories; no single donor follows a comparable pre-treatment path, which is precisely why DiD identification is invalid for this category and synthetic control is the appropriate method.

Figure 6 displays the actual and synthetic Food series. The gap between actual and synthetic Food is positive in the first post-treatment year ( $t = 2024$ ), reaching +0.993 log points, with a post/pre RMSPE ratio of 1.39. This positive gap is the RSA’s estimated effect on food waste recycling following the mandatory segregation requirements introduced in March 2024: actual Food recycling outperforms its synthetic counterfactual by approximately 170% in percentage terms ( $e^{0.993} - 1 \approx 1.70$ ) in the first post-treatment year. The gap magnitude is notable, though the post/pre RMSPE ratio of 1.39 is modest by the standards of synthetic control studies, reflecting the difficulty of constructing a close synthetic counterfactual for Food. However, the estimate rests on a single post-treatment year, which makes it impossible to distinguish a genuine policy effect from a one-year anomaly. A reliable causal assessment requires at least three post-treatment years of data; the current result establishes a directional finding to be confirmed as data accumulate beyond 2024.

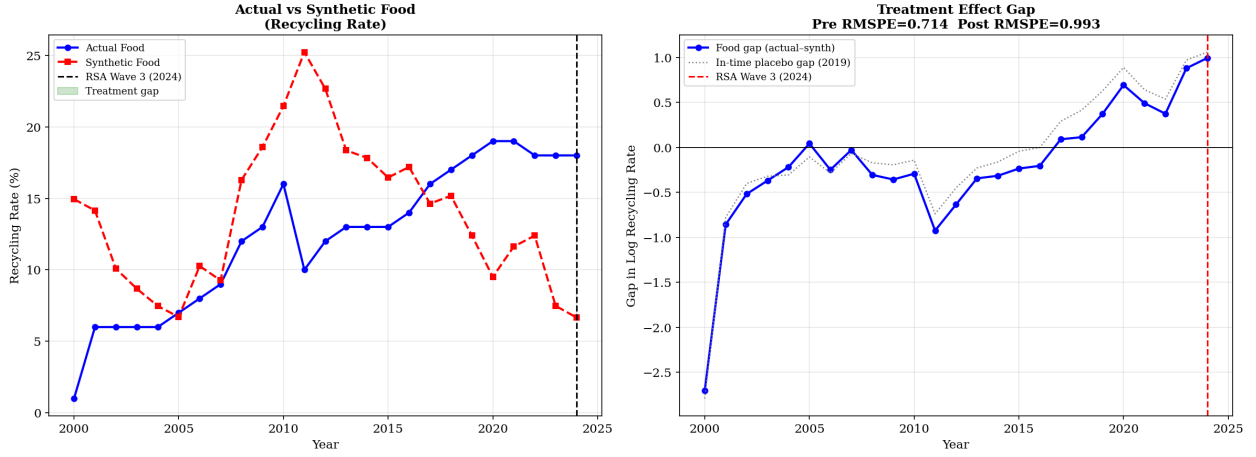


Figure 6: Synthetic Control: Food Waste Recycling Rate (2000-2024).

*Notes:* *Left panel:* Y-axis: recycling rate (%); X-axis: calendar year, 2000-2024. Blue solid line: actual Food recycling rate; red dashed line: synthetic Food counterfactual, constructed as a weighted combination of Glass (85.6%) and Plastic (14.4%). The vertical dashed line marks RSA Wave 3 (2024). *Right panel:* Y-axis: treatment gap in log recycling rate units; X-axis: year. The grey dotted line is an in-time placebo gap using a fictitious 2019 treatment date. Pre-treatment RMSPE = 0.714 log points; post-treatment RMSPE = 0.993 log points (ratio 1.39).

## 5 Conclusion

This paper estimates the causal effect of Singapore’s Resource Sustainability Act on recycling rates using a staggered DiD framework with two-way fixed effects and cluster-robust standard errors. Wave 1 packaging categories (Paper, Plastic, Glass) experienced a significant decline of 0.47 log points relative to never-treated controls following the 2020 RSA rollout, an effect that remains stable across COVID-19 robustness checks. Wave 2 metal categories show no measurable response, which is expected: categories already recycling at 99% have no behavioural margin to adjust re-

gardless of regulatory pressure. The event study reveals that the Wave 1 decline deepens over four years, reaching  $-0.2807$  log points at  $t + 4$ , consistent with firms progressively underestimating the infrastructure and reporting demands of the new compliance regime.

Callaway-Sant’Anna group-time estimates confirm this trajectory is not an artefact of TWFE negative weighting, strengthening the causal interpretation. For food waste, the synthetic control finds a notable positive gap of  $+0.993$  log points in the first post-treatment year, suggesting Wave 3 mandatory segregation requirements improved outcomes, though this should be interpreted as a directional finding only.

Future work should extend the panel beyond 2024, incorporate firm-level compliance records to identify which producers drove the packaging decline, and implement the full Callaway-Sant’Anna estimator for inference-valid standard errors on the group-time ATT estimates.

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## A Appendix

### A.1 Correlation Matrix

Figure 7 presents pairwise Pearson correlations between RSA-targeted and control categories over the pre-treatment period (2000-2019). Packaging categories (Paper, Glass) show moderate positive correlations with controls ( $r \approx 0.18$ - $0.21$ ), while Plastic shows a strong negative correlation with both controls ( $r = -0.71$  for Construction & Demolition,  $r = -0.78$  for Used Slag). This asymmetry motivates the log transformation applied in the main analysis. Food shows high positive correlation with controls ( $r = 0.66$ - $0.69$ ) but with a substantially steeper slope, motivating the synthetic control approach for Wave 3.

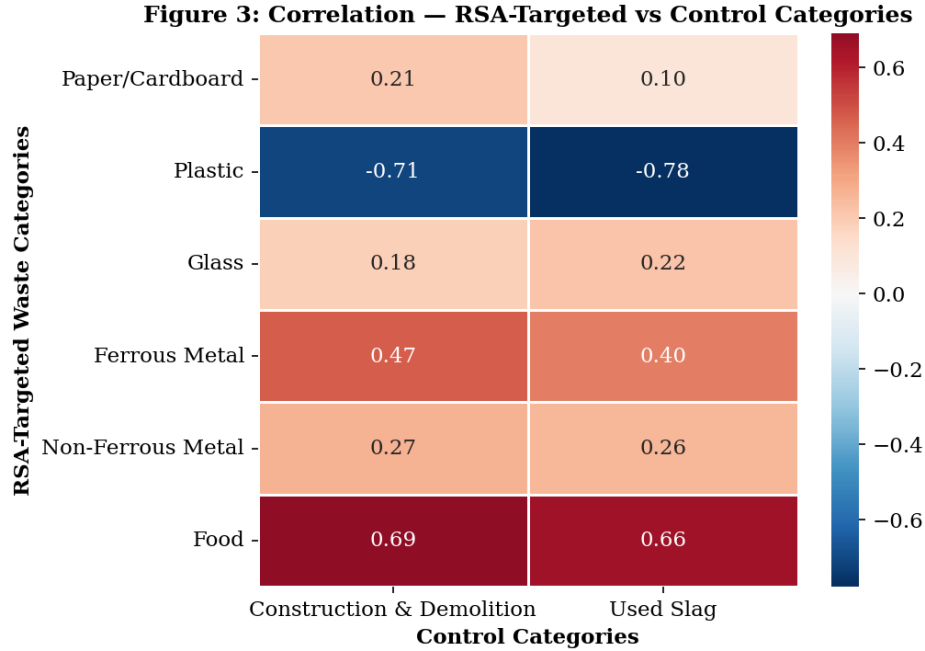


Figure 7: Correlation Matrix: RSA-Targeted vs. Control Categories (2000-2019).

*Notes:* Heatmap of pairwise Pearson correlations between annual recycling rates. Rows: RSA-targeted categories; columns: never-treated controls (Construction & Demolition, Used Slag). Sample: pre-treatment period, 2000-2019.

Colour scale: deep blue = strong negative correlation ( $r$  near  $-1$ ); deep red = strong positive correlation ( $r$  near  $+1$ ). Each cell shows the  $r$  value between that row category and column control.

## A.2 Waste Composition Trends

Figure 8 decomposes total annual waste (thousands of tonnes) into recycled and disposed components from 2000 to 2024. Total waste generation peaked around 2012-2013 at approximately 6,400 thousand tonnes and in the year 2020, the waste volume was the lowest. The stable recycled share over the full period masks substantial heterogeneity across categories, as documented in Table 2.

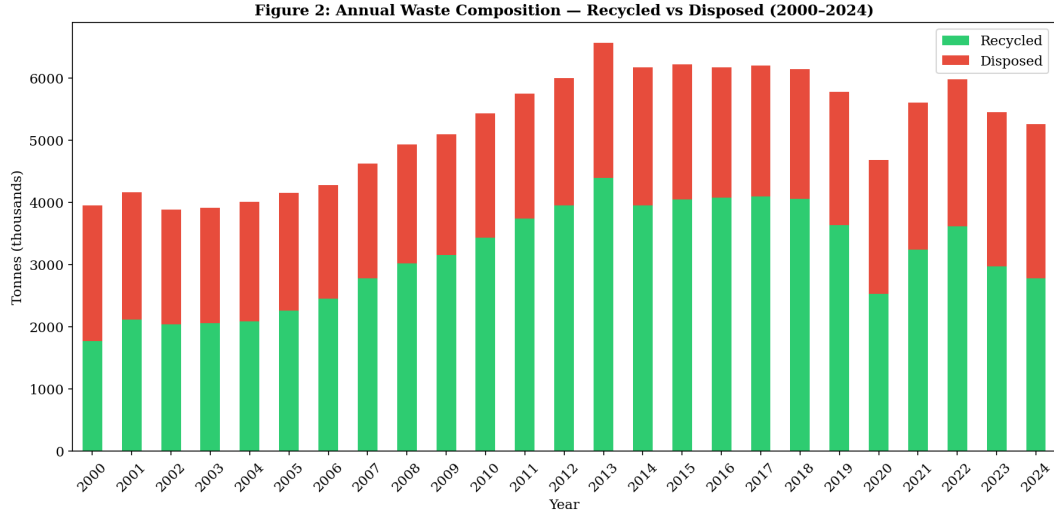


Figure 8: Annual Waste Composition: Recycled vs. Disposed (2000-2024).

*Notes:* Y-axis: total annual waste (thousands of tonnes). X-axis: Years 2000-2024. Each bar represents a year, stacked into recycled (green) and disposed (red) components across all eight retained waste categories.